TeraLyzer Manual

Lytera UG (haftungsbeschränkt)

Revision 1.3

Dear Customer,

Thank you very much for purchasing the TeraLyzer Data Extraction Software. On the following pages we will introduce you to the features of the Software and will discuss each step in the data evaluation process. Aside from this manual please also have a look on our homepage www.lytera.com for recent case studies that we will post there on a regular basis.

If you have any questions regarding the software or if you should encounter any problems while processing one of your own datasets, please do not hesitate to contact our support via email: support@lytera.com

We will do our best to make your user experience most satisfying and are glad for any suggestions on improvements of the TeraLyzer Software.

Sincerely yours,

The LyTera Team

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1 General Remarks

Before we delve into the depths of the data evaluation, we shall consider the conditions under which the TeraLyzer can operate. The TeraLyzer was written for one specific purpose:

- **Extracting highly precise material parameters (n, α, κ, ε',ε'')**
- **from transmission terahertz time domain spectroscopy data**
- **complete with error bars**
- **from both single- and multilayer samples.**

The TeraLyzer software has been extensively tested with data from various terahertz (THz) time domain spectroscopy (TDS) setups. Emitter schemes included photoconductive antennas, surface emitters, optical rectification and generation with multimode laser diodes. As detectors, both electro optic sampling and photoconductive antennas were tested. In all these setups, the data extraction worked smoothly.

Please note, that at the current development stage, the TeraLyzer software is only suited for the evaluation of data sets obtained from a transmission experiment and can neither extract data from reflection experiments nor from measurements obtained in differential THz TDS setups.

In section 0 you will find hands-on information on how to handle the TeraLyzer. The important measurement parameters required for extracting reliable material parameters are highlighted and discussed. Please read this section carefully to avoid undesired confusion in interpreting the resulting data.

2 Theoretical Background

Many different procedures for the analysis of THz TDS data have been proposed in literature [1, 2, 3, 4]. While the TeraLyzer combines multiple aspects from these sources, reference [1] contains the information about the specific algorithm employed in the software. The inclined reader is referred to this reference for further information.

Before the material parameter extraction can be applied, a reference pulse with an empty THz TDS system and a sample pulse with the sample in the THz beam path are measured. The ratio of the corresponding Fourier spectra yields the measured transfer function $H_{meas}(\omega)$. The further procedure depends on the nature of the. Both, single and multilayer samples can be investigated where in case of multilayer samples the material parameters of M-1 layers in an M layer stack have to be known in advance.

2.1 Single Layer Data Extraction

In case of a single layer sample, the theoretical transfer function takes the form of

$$H_{theory,single}(\omega) = (1-r^2)e^{-i(\tilde{n}-1)k_0\Delta} \cdot \sum_{m=0}^{M}\left(r^2 e^{-i2\tilde{n}k_0\Delta}\right)^m \qquad (1)$$

[1] M. Scheller, C. Jansen, and M. Koch, “Analyzing sub-100-µm samples with transmission terahertz time domain spectroscopy”, *Optics Communications*, vol. 282, no. 7, pp. 1304–1306, 2009.

[2] T.D. Dorney, R.G. Baraniuk, and D.M. Mittleman, “Material parameter estimation with terahertz time-domain spectroscopy”, J. Opt. Soc. Am. A, vol. 18, no. 7, pp. 1562–1571, 2001.

[3] L. Duvillaret, F. Garet, and J.-L. Coutaz, “Highly precise determination of optical constants and sample thickness in terahertz time-domain spectroscopy”, Appl. Opt., vol. 38, no. 2, pp. 409–415, 1999.

[4] I. Pupeza, R. Wilk, and M. Koch, “Highly accurate optical material parameter determination with THz time-domain spectroscopy”, *Opt. Express*, vol. 15, no. 7, pp. 4335–4350, 2007.

with *r* as Fresnel reflection coefficient, k_0 as the free-space wave number, c_0 as the speed of light in vacuum, Δ as the thickness of the sample, $\tilde{n} = n - i\kappa$ as the complex index of refraction and *m* as the number of echo pulses included in the corresponding measured transfer function accounting for the limited length of the measurement time window. By minimizing the difference between the measured and the theoretical transfer function, $H_{meas}(\omega) - H_{theory,single}(\omega)$, using an implementation of the Nelder-Mead simplex algorithm as described by Lagarias et al. in [5], the thickness- and frequency-dependent complex refractive index of the sample is calculated in a numerical optimization.

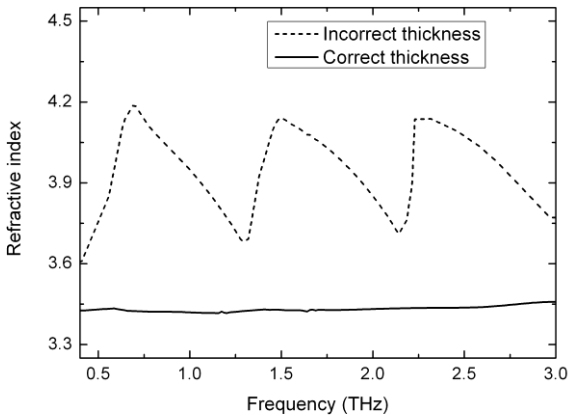


Fig. 1: The extracted refractive index for two different thicknesses.

To determine the sample thickness, the Fabry-Pérot (FP) oscillations superimposed on the refractive index data are analyzed. These oscillations result from FP echo pulses included in the time window of the measurement. If the correct thickness is chosen, these oscillations are supposed to disappear [1]. As an example, the extracted refractive index of Polypropylene is shown in the figure 1 for two different assumed thicknesses. The figure illustrates how the FP oscillations compromise the material data when a

[5] Jeffrey C. Lagarias, James A. Reeds, Margaret H. Wright, and Paul E. Wright, "Convergence properties of the nelder–mead simplex method in low dimensions", *SIAM J. on Optimization*, vol. 9, no. 1, pp. 112–147, 1998.

wrong thickness is considered. Thus, a selective investigation of these oscillations allows for an accurate thickness determination.

For analyzing the FP oscillations two approaches are employed: The first one is the quasi-space (QS) method [1]. Here, a Fourier transform of the frequency dependent refractive index and the frequency dependent extinction coefficient is performed so that the sinusoidal FP oscillations result in a distinct peak. By minimizing this peak, the correct thickness is determined and the correct complex refractive index of the sample is obtained.

As second approach, the total-variation (TV) method [4] is employed. The TV denotes the total variation of the neighboring values of the refractive index and the extinction coefficient. The FP-oscillations will result in an increased TV value. However, since all variations are considered, also absorption peaks and noise induced features will result in a high TV. Thus, in most cases the QS-approach is the method of choice for analyzing the thickness, especially for samples that are thinner than 500 µm as the TV method starts to fail for very thin samples. For relatively thick samples in the range of several tens of mm, the TV approach might perform better though. The reason for that is that the TV method also works if the FP-oscillation period is close to the frequency resolution of the measurement.

2.2 Multi Layer Data Extraction

Now that the material parameters of the single layers are available, the data extraction algorithm proposed in [1] can be extended so that multilayer data can be extracted. While the basic procedure remains unchanged, the theoretical transfer function now has to represent a stratified medium including all multiple reflections at the interfaces occurring inside the sample.

The TeraLyzer software offers two different approaches to create a theoretical multilayer transfer function: A Transfer Matrix and a Ray Tracing formulation. While the transfer matrix approach offers a much higher computational efficiency, it inherently assumes that all internal reflections fall within the measured time window. In contrast to that, the Ray Tracing approach accounts for the finite time window but is relatively complex and computation intensive.

Representatively, we will discuss the transfer matrix formulation here. In the transfer matrix approach the propagation of an electromagnetic wave through a multi layer medium is described by a multiplication of characteristic propagation and interface matrices P_i and I_i of the individual layers:

$$\begin{pmatrix} E_{inc} \\ E_{refl} \end{pmatrix} = I_0 \left(\prod_{i=1}^{M} \mathrm{P}_i I_i \right) \begin{pmatrix} E_{trans} \\ 0 \end{pmatrix} = \begin{pmatrix} \xi_{11} \\ \xi_{21} \end{pmatrix} E_{trans} \qquad (2)$$

Here, P_i constitutes the propagation through the ith layer and I_i considers the interaction of the wave with the interface between the layer *i* and the layer *i+1*. P_i and I_i are given by

$$P_i = \begin{pmatrix} e^{i\tilde{n}_i k_0 \Delta_i} & 0 \\ 0 & e^{-i\tilde{n}_i k_0 \Delta_i} \end{pmatrix} \qquad (3)$$

and

$$I_i = \frac{1}{t_{i,i+1}} \begin{pmatrix} 1 & r_{i,i+1} \\ r_{i,i+1} & 1 \end{pmatrix} \qquad (4)$$

with Δ_i as the thickness and n_i as the complex refractive index of the ith layer, $r_{i,i+1}$ and $t_{i,i+1}$ as the Fresnel reflection and transmission coefficient between the layer *i* and the layer *i+1* and k_0 as the free-space wave number. From equation (2), the theoretical transfer function can be derived, which refers to the transmission with the sample in place normalized to the transmission through air:

$$H_{theory,multi}(\omega) = \frac{E_{trans}}{E_{inc} e^{-ik_0 \sum \Delta_i}} = \frac{\frac{E_{inc}}{\xi_{11}}}{E_{inc} e^{-ik_0 \sum \Delta_i}} = \left(\xi_{11} e^{-ik_0 \sum \Delta_i} \right)^{-1} \qquad (5)$$

Now that the theoretical multilayer transfer function is defined, the same steps as discussed above are performed using the transfer function of the multilayer stack. After calculating the complex refractive index of the investigated layer as a function of the frequency and the thickness by numerical optimization, the quasi space analysis unambiguously reveals the thickness of the sample layer.

2.3 Important Aspects for the Data Analysis

As described above, the FP echo pulses are essential for employing the thickness determination feature of the TeraLyzer software. Be careful that the chosen time window of your measurement includes at least one of these echo pulses. You can estimate the necessary length of the time window by considering the time difference between the FP echo and the pulse that propagates directly through the sample:

$$\Delta T_{FP} = \frac{2\,n\Delta}{c_0}. \qquad (6)$$

Keep in mind that the first FP echo is delayed by ΔT_{FP} in respect to the directly transmitted sample pulse. A too short time window will compromise the thickness extraction process. As rule of thumb, it is advisable to choose a time window that includes at least two FP echoes. In this case, the frequency resolution of the measurement will be sufficient to enables for a reliable thickness determination.

The second important parameter is the bandwidth of the measurement. The time difference between the FP echoes is the inverse of the FP oscillation period ΔF_{FP}:

$$\Delta F_{FP} = \frac{c_0}{2\,n\Delta}. \qquad (7)$$

To analyze the perturbing FP oscillations, it is necessary that the extracted material parameters comprise at least a single complete FP period. That means the bandwidth of the measurement has to be larger than ΔF_{FP}.

Example: For analyzing a sample with an optical thickness (refractive index times geometrical thickness) of 100 µm the material parameters have to be extracted over a frequency interval of at least 3 THz. Please note that the signal-to-noise ratio within this interval has to be high enough to extract reliable material parameter.

Figure 2 illustrates this requirement. A thin silicon wafer was measured with a spectrometer that exhibits a bandwidth of only 1 THz. As can be seen in the figure, the extracted refractive index doesn't comprise a complete FP oscillation before the noise impedes a reliable extraction of the material parameters. Thus, the thickness determination is obstructed.

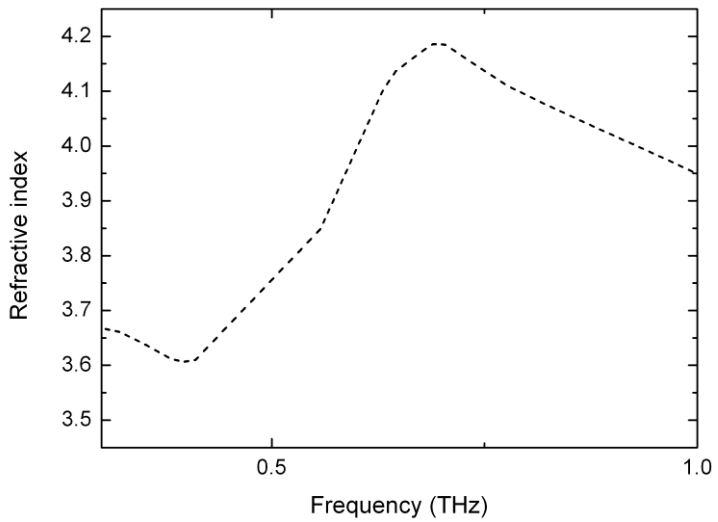


Fig. 2: The extracted refractive index in the case of insufficient bandwidth.

2.4 SVMAF Filtering

As Aristotle said:

"It is the mark of an educated man to look for precision in each class of things just so far as the nature of the subject admits."

Correspondingly, it is essential to notice that the extracted material parameters are just valid in the range of the confidence interval of the measurement. That means within this interval there is no evidential truth which value is the real physical one but depending on the chosen width of the confidence interval there is a given probability that the true value lies within its borders. Thus, it is legitimate to eliminate variations in the material parameters attributed to the influence of noise by the SVMAF filtering method (Spatially Moving Average Filter) as further depicted in [4] as long as the resulting values all remain in the previously calculated confidence interval.

3 Data Extraction in Four Steps

The TeraLyzer user interface consists of four major windows which take you through the data extraction process

- **The Data Import Window**
- **The Time Domain Preprocessing Window**
- **The Solver Window**
- **The Result Presentation and Data Export Window**

First, the THz time domain data is loaded from ASCII files in the data import window. Multiple data sets can be used so that a basis for the statistical evaluation is provided and reliable error bars can be calculated during a later step in the data evaluation.

Secondly, the Time Domain Preprocessing Window is opened where the time domain THz traces are optimized for the following frequency domain analysis, including steps such as windowing, DC offset removal, zero padding, etc.

Thridly, the solver window is displayed, giving some information about the progress of the numerical optimization.

Finally, the Result Presentation window pops up, where the material parameters of the sample and the results of the thickness determination are displayed.

In the remainder of this section, we will go through each of this four steps and give information about the single options.

3.1 Step I – Data Import

The TeraLyzer imports THz time domain data from ASCII files. The files can contain additional information in form of header rows. **At least one time and one data column has to be present. Each column has to be separated by a TAB.** The Import Options Panel allows for various settings.

Please note that only one sample can be characterized at a time, so only load the data sets corresponding to a single sample.

Time Column

Selects the column (starting from the left) which contains the time vector.

Data Column

Selects the columns which contain the time domain terahertz signal amplitude data. Multiple columns can be loaded simultaneously. The first field selects the first data column and the second field the last data column which should be imported

File Type

Only display files of selected file type in the open file dialog which appears when you hit the Add button. Default is *.* which shows all files.

Normalize Time By

The time vector is divided by this value. If Normalize Time By is set to 1, then the TeraLyzer assumes that the time in the file is specified in seconds. If it is set to 1e12, the time in the file should be picoseconds, in case of 1e15 the time should be femtoseconds, and so on.

This option is also helpful if you should have saved the delayline position e.g. in µm instead of a time vector.

In this case, you can chose a proper normalization factor using the speed of light to transform the vector to a time vector.

Number of Header Rows

Select the number of header rows which do not contain the time or THz amplitude data.

Resample Data

If this option is selected, the time domain data is resampled to the given sampling time. If this option is not selected but non-equidistant time vector is loaded, then a resampling to the smallest time increment in the time vector is performed. In case of an equidistant time vector, no resampling is performed unless this option is explicitly selected. Please note, that in contrast to the other parameters which can be selected in the Import Option Panel, the resampling does not immediately take place upon loading the data sets but only after continuing to the Time Domain Processing window.

Add Buttons

By clicking the respective Add Button, reference or sample data sets can be loaded. In the open file dialog, multiple files can be selected at once. Doing so applies the import options provided in the Import Options Panel to all of these files.

Remove Buttons

By first selecting a data set either via a click into the plot window or into the listbox with the data set names and then hitting the Remove Button, the corresponding data is removed from the evaluation.

Zoom-In/ Zoom-Out Buttons

In the top of the program window, a Zoom-In and a Zoom-Out Button are present. By clicking one of these the plotted curves can be zoomed in for a more detailed view. When the zoom is selected, a double click on the plot window restores the original view.

Data Cursor Button

Besides the Zoom Buttons, the tray bar in the top section of the program view also contains a data cursor. By selecting the Data Cursor Button and then clicking on one of the THz traces, the corresponding x- and y-values are displayed. The cursor is locked to the trace and can be moved in a drag and drop style.

Next Button (>>)

When all data sets are loaded as desired, the Next Button closes the Data Import and opens the Time Domain Preprocessing Window.

3.2 Step II – Time Domain Preprocessing

Now that the time domain data is loaded into the program, the Time Domain Preprocessing Window offers a wide range of options to optimally condition the data sets for the following analysis, which can be accessed from the Signal Processing Panel.

Three plots show the effect of the applied changes: A time domain plot of the amplitude data (left), a frequency domain plot of the phase difference between reference and sample (middle) and a frequency domain plot with the amplitude spectra of reference and sample (right).

After the time domain data is conditioned as suited, the parameters for the following evaluation process can be selected in the Evaluation Settings Panel. When all the settings are performed the Optimization Window is opened.

3.2.1 Signal Processing Panel

Start Time (sample), End Time (reference) and Windowing

Select the time window which you would like to consider. To do so, you select the start time from the sample pulse and the end time from the reference and whether you would like to use a Blackman window or no window function (corresponds to a rectangular window).

The Blackman window provides a soft roll-off at the outer limits of your time window, which can proof beneficial as sharp steps are avoided. If you decide to use a Blackman window, you can specify the window slope (the roll-off time span in which the window has a value other than unity).

The windows are applied symmetrically. Thus the sample window is shifted with regards to the reference window by the same amount of time that the pulse maximum position of the sample pulse differs from that of the reference pulse.

When selecting the time range, please keep in mind, that in order to extract the thickness information simultaneously with the material parameters, you will have to include at least one Fabry Perot Echo pulse (also see section 2.4). The TeraLyzer will also perform an automatic check and inform you if that echo pulse should not be present in the time window you specified as soon as you hit the Analyze Sample Button.

Furthermore, the length of the time window determines the obtainable frequency resolution. This aspect might not be crucial for most samples, however, if you are investigating sharp resonances, the time window should be selected appropriately.

Offset Compensation

To avoid a DC component on the terahertz spectra, this option should be used to remove any constant offsets from the time domain data. It takes the specified time from the beginning of the time window and averages the amplitude values. This average is then subtracted from the amplitude data.

Zero adding

If desired, the TeraLyzer can add zeros to your time domain data. Doing so artificially enhances the frequency resolution. However, not physical relevant information is added, so that this enhancement should be treated very carefully. Still, some cases exist where this option can be very helpful.

Phase Offset Removal

As the THz TDS signal typically does not extent down to 0 Hz, the phase in the low frequency regime is distorted. Using the Phase Offset Removal Option, you can select a frequency interval in which the phase information is in good shape. From this interval, a linear interpolation down to 0 Hz is performed and 2π are added or subtracted until the 0 Hz value of the phase difference lies within the $0 \pm 2\pi$ window.

Noise starts at

In this option, you should select a frequency from which on the data is purely noise. On basis of this noise floor, the dynamic range and the signal to noise ratio are determined. Later in the data evaluation process, this information will be used to determine the valid frequency range for the extracted material parameters from the dynamic range as described by Jepsen et al. in [6].

Apply Button

The Apply Button puts the settings made in the Signal Processing Panel into effect and updates the time and frequency domain plots.

[6] P. U. Jepsen and B. M. Fischer, "Dynamic range in terahertz time-domain transmission and reflection spectroscopy", Optics Letters, vol. 30, no. 1, pp. 29–31, 2005.

3.2.2 Evaluation Settings Panel

Thickness

Here, the approximate sample thickness together with an interval around it and a step size are specified. For example, if the object under investigation has a thickness of approximately 1 mm ±50 µm and you would like to know the thickness with a precision of 5 µm then you would insert these values in the corresponding fields.

Please keep in mind, that if you select a broad interval and a small step size, the evaluation time might become very large. If you have to consider a large interval, then a good approach is to evaluate this large interval with a coarse stepping and then redo the evaluation in a smaller interval with a fine stepping.

Extract Material Parameters (frequency range)

Select the frequency rage in which you would like to extract the material parameters. The values given below the fields are suggestions only. The larger the frequency interval which you are interested in the longer the evaluation will take. Please keep this tradeoff in mind.

3.2.3 Multilayer Setup Button

If the sample is a multilayer structure, the multilayer setup can be evoked by hitting the Multilayer Setup Button.

In order to analyze a single layer inside a multilayer stack all other layers have to be already characterized in corresponding single layer scans.

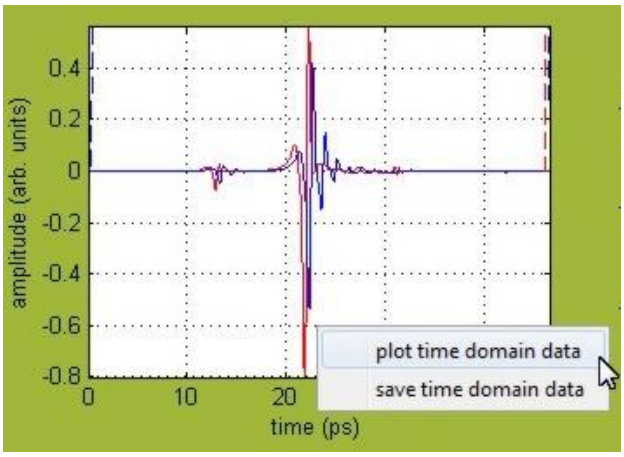
The files exported from these single layer scans can now be loaded by hitting the Get Transmission File Button. By using the Add Layer and Remove Layer Buttons, new layers can be added and subtracted from the stack. The Thickness can be manually set to other values than saved in the layer file. The unidentified layer can be added with the Add Investigated Layer Button.

The material parameters of the single layers can be either loaded in form of the SVMAF data (see section 0) or the unfiltered data. The wave propagation direction through the stack is marked by the arrows on the right hand side (from the bottom to the top). When the multilayer settings are performed hit the OK button.

3.2.4 Other Options

Right-Click Plot Menus

After the Apply Button is clicked you can access data export menus by right clicking the time domain plots or the frequency domain spectra.



In these menus you can select to plot the time domain traces, the amplitude spectra or the transfer function in a separate plot window so that you can save the data as images. Furthermore, you can export the time domain or frequency domain data to comma separated files for archiving or for further processing outside the TeraLyzer.

Zoom In/ Zoom Out Buttons

In the top of the program window, a Zoom In and a Zoom Out Button are present. By clicking one of these, the plotted curves can be zoomed in for a more detailed view. When the zoom is selected, a double click on the plot window restores the original view.

Data Cursor Button

Besides the Zoom Buttons, the tray bar in the top section of the program view also contains a data cursor. By selecting the Data Cursor Button and then clicking on one of the curves, the corresponding x- and y-values are displayed. The cursor is locked to the trace and can be moved in a drag and drop style.

Save and Load Parameter Set Buttons

In the tray bar are also a load and a save button. These buttons can be used to save or restore a parameter set with all settings made in the Data Import Window and in the Time Domain Preprocessing Window.

User assumed number of pulses

The TeraLyzer automatically determines based on your input information how many Fabry Perot echo pulses fall within the evaluated time domain window. If you would like to assume another number instead, you can specify this number here. This is an advanced user's option and should only be changed if the specific scenario requires it.

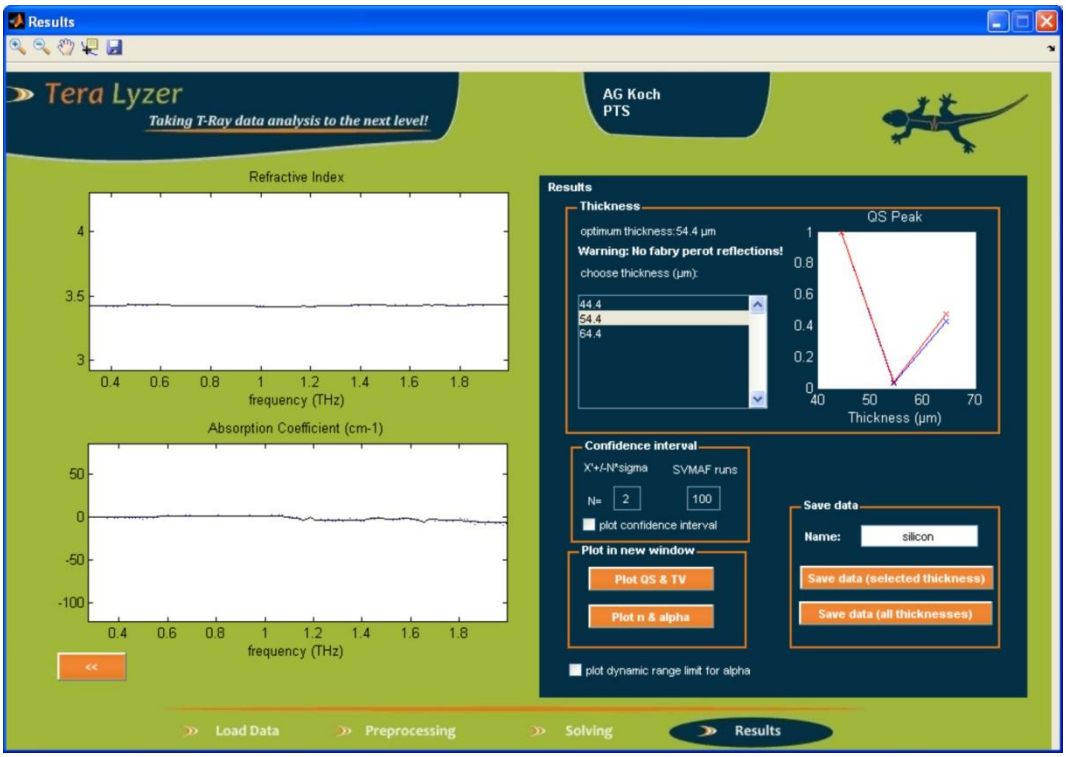
3.3 Step III – Numerical Optimization

After all settings have been chosen, the data extraction can begin. The third window guides you through the numerical data extraction process. Before the extraction begins, information about the complexity of the computation is provided. The numbers of frequency steps as well as of thickness steps are shown in the upper right of the figure. The calculation time scales linear with the product of both values.

If a multilayer sample is investigated, you may choose between a fast and an accurate approach for calculation the material parameters. The accurate method bases on a ray tracing scheme to calculate the FP echoes that are included in the time window of the measurement. The fast method bases on a transfer matrix solver and assumes an infinite long time window of the measurement, which reduces the calculation afford but may compromise the extracted data in certain cases. Thus, only in the case of a sufficiently long measurement time window the fast approach is recommended. For more information please also read section 2.2.

The Start Calculation button initiates the computation process. For each thickness and each frequency step the material parameter are successively extracted. The actual values are presented on the left hand side of the window. On the right side a pie chart illustrates the progress of the computation. Once the extraction is finished, the result window will be displayed.

3.4 Step IV - Display of Results and Data Export



In the final step the extracted material parameter, the refractive index and the absorption coefficient are visualized. In addition, the result of the thickness determination is presented on the right hand side of the window. The QS-value is a figure of merit for the strength of the FP-oscillations superimposed to the material parameter. For the correct thickness, these oscillations vanish and the QS curve reaches its minimum value. Yet, if an inappropriate thickness interval was not chosen or no FP-echoes were included in the time window of the measurement, the QS values will not exhibit a V-shaped curve. This is an indication that some error might have occurred

during the data extraction. Please check carefully, that all the prerequisites named in section 0 are fulfilled.

You may also see the material parameters of each analyzed thickness. For that, simply select one of the listed thickness value in the list box **thickness**.

To visualize the error bars of your measurement, activate the **error bars** checkbox. The plotted curves of the material parameters will then be enclosed by the confidence interval of the measurement. Please note that this interval is only of physical sense if you have analyzed a data set that consists at least of 2 reference measurements and 2 sample measurements! Otherwise this interval is just a mathematical description of your waveform's signal-to-noise ration that doesn't take any statistics into account. By changing the value of N you increase your confidence interval according to [4]. The number of SVMAF runs denotes the filter strength of the SVMA-Filter as further described in [4].

If you would like to have a closer look to the material parameters press the **Plot n & alpha** button. A new figure will open that allows for an extended view onto these values. Same goes for the **Plot QS & TV parameter** button. This will open figures that contain information about the FP-oscillations necessary for the thickness determination. On the one hand you will see an enlarged view of the QS value. On the other hand, the TV value is illustrated both as absolute values and as histogram. Please note that the QS predication is mostly superior to the TV value.

You can save the material parameters for the selected thickness by pressing the **save data (selected thickness)** bottom. In this case, two files with the selected filename will be generated. One contains the data as comma-separated-values (csv), the other one as Matlab® variables in a Matlab® compatible .mat archive.

The files (csv and mat) contain the following parameters:

Parameter name (csv), Parameter name (mat)
Description
freq (csv), freq (mat):
The frequency array
ref_ind (csv), nModel (mat):
The extracted refractive index
kappa (csv), kModel (mat):
The extracted extinction coefficient
alpha (csv), aModel (mat):

The extracted absorption coefficient
epsilon_r (csv):
The extracted real part of the permittivity
epsilon_i (csv):
The extracted imaginary part of the permittivity
epsModel (mat):
The extracted complex permittivity
ref_ind_SMUV (csv), nSMUV (mat):
The extracted SVMA-filtered refractive index
kappa_SMUV (csv), kSMUV (mat):
The extracted SVMA-filtered extinction coefficient
alpha_SMUV (csv), aSMUV (mat):
The extracted SVMA-filtered absorption coefficient
epsilon_r_SMUV (csv):
The extracted SVMA-filtered real part of the permittivity
epsilon_i_SMUV (csv):
The extracted SVMA-filtered imaginary part of the permittivity
epsModel (mat):
The extracted SVMA-filtered complex permittivity
alpha_max (csv), alpha_max (mat):
The value of alpha_Max according to [6]
delta_N (csv), NsigN(mat):
The error bars for the refractive index
delta_A (csv), NsigA(mat):
The error bars for the absorption coefficient
H_measure_r (csv),:
The measured real part of the transfer function
H_measure_i (csv),:
The measured imaginary part of the transfer function
Hmess (mat):
The measured complex transfer function
H_calc_r (csv),:
The calculated real part of the transfer function based on the extracted parameters
H_calc_i (csv),:
The calculated imaginary part of the transfer function based on the extracted parameters
HsimOut (mat):
The calculated complex transfer function based on the extracted parameters
T_measure (csv), T_mess (mat):
The measured absolute value of the transfer function
T_calc (csv) , T_simOut (mat):

The calculated absolute value of the transfer function based on the extracted parameters
Thickness (csv),:
The selected thickness
allThicknessess (csv),:
Array of the investigated thicknesses
QSValue (csv), QSpeak (mat):
The corresponding QS-Values of the investigated thicknesses based on the refractive index
KSpeak (csv), KSpeak (mat):
The corresponding QS-Values of the investigated thicknesses based on the extinction coefficient
Num_Thickness (csv), N_thick (mat):
A figure of merit for the prediction of the TV based evaluation. A relative high number at a single thickness means convergences of the TV method.
N_std (csv), N_sig (mat):
The value of N used to calculate the error bars

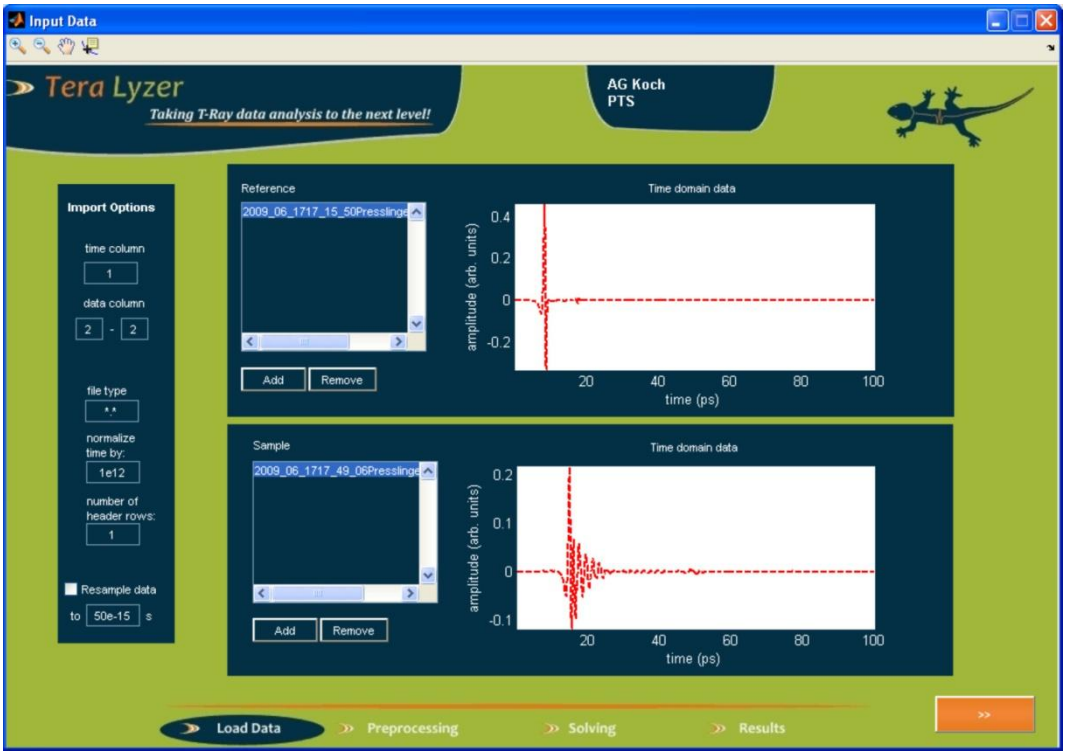
To save the parameters for all extracted thicknesses press the **save data (all thicknesses)** bottom. In this case, a Matlab® compatible .mat archive is generated that contains all the extracted values, but no CSV version will be saved as the arrays would grow to large. Please select a single thickness for CSV export.

4 Case Studies

4.1 Investigating Sharp Absorption Peaks in α-lactose

4.1.1 Step I - Loading the time domain data

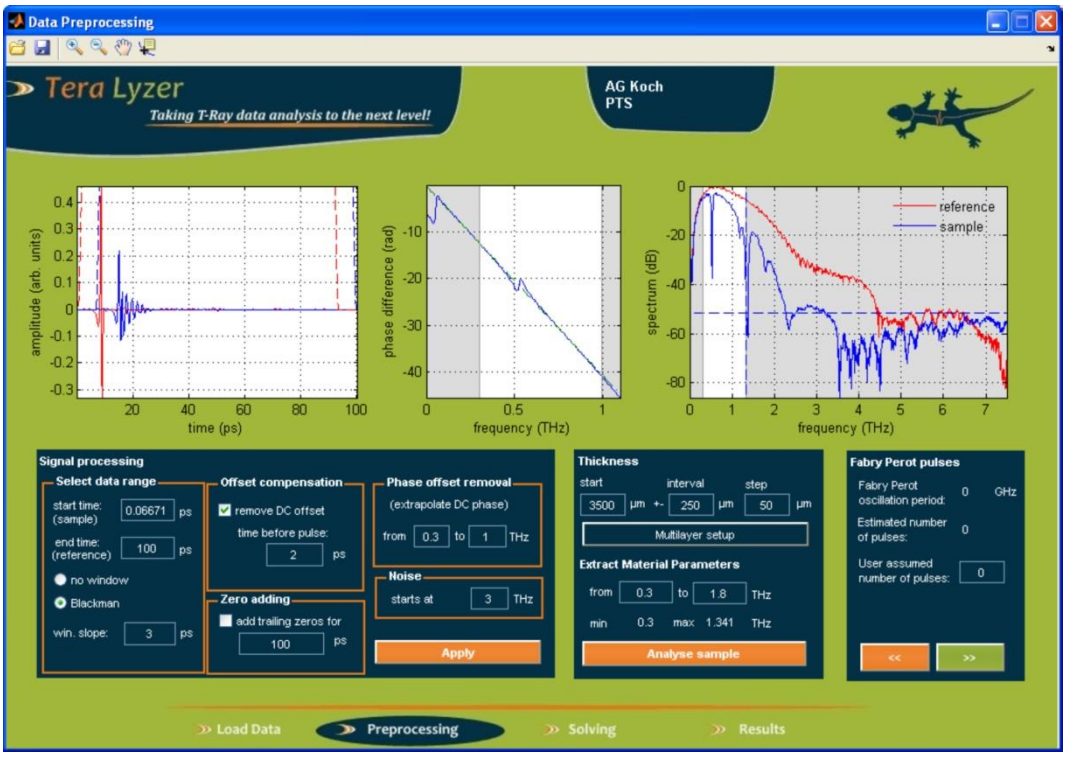
Import your recorded THz time domain pulses from an ASCII file into the TeraLyzer. The import window helps you to find the correct import settings. For advanced statistics, you can load multiple data sets.



In our case, the ASCII file's first column contains the time in ps and the second tab-separated column contains the corresponding THz data. Thus we set the "time column" to 1 and the "data column" to 2. As the time axis has the unit ps we set the "normalize time by" field to 1e12 to obtain the time axis in seconds. Our data has no header rows so that we leave the "header row" option 0.

4.1.2 Step II - Preprocessing

The next window offers many preprocessing options like DC offset removal, time-domain windowin, phase correction and appending zeros. Furthermore, the frequency range for which parameters should be extracted can be determined here.



In our case, we select the full available data range from 0 to 100 ps and apply a Blackman window with a slope of 3 ps. The DC offset is removed by averaging over the first 2 ps of the time domain data and subtracting this average from the data set. As phase errors may occur at very low frequencies due to non-planar phase fronts and strong diffraction, we select a trusted phase regime from 0.3 to 1 THz. From the spectral data, we see that the noise starts at roughly 2.2 THz, so we chose this value and enter it into the program. By hitting the "Apply" button, the changes take place.

Next, we chose the sample thickness to be within 3500 µm +/-250 µm and the iteration step size to 5µm. The material parameter extraction frequency range is set to 0.3 THz to 1.8 THz. By hitting the "Analyze Sample" button, an initial evaluation is performed and the position of the Fabry Perot echoes is displayed in the time domain data plot. We see that 2 Fabry Perot echoes lie within our sample range.

4.1.3 Step III - Numerical Material Parameter Extraction

After the Preprocessing is performed, the calculation of the parameters takes place. Depending on the thickness iteration steps, this process might take a few minutes.

In the following window, the material parameters are displayed.

4.1.4 Step IV - Material Parameters Results

After the calculation is finished, the resulting material parameters are displayed. The well-known absorption peaks of alpha lactose compound are well reproduced and respective features are found in the refractive index.

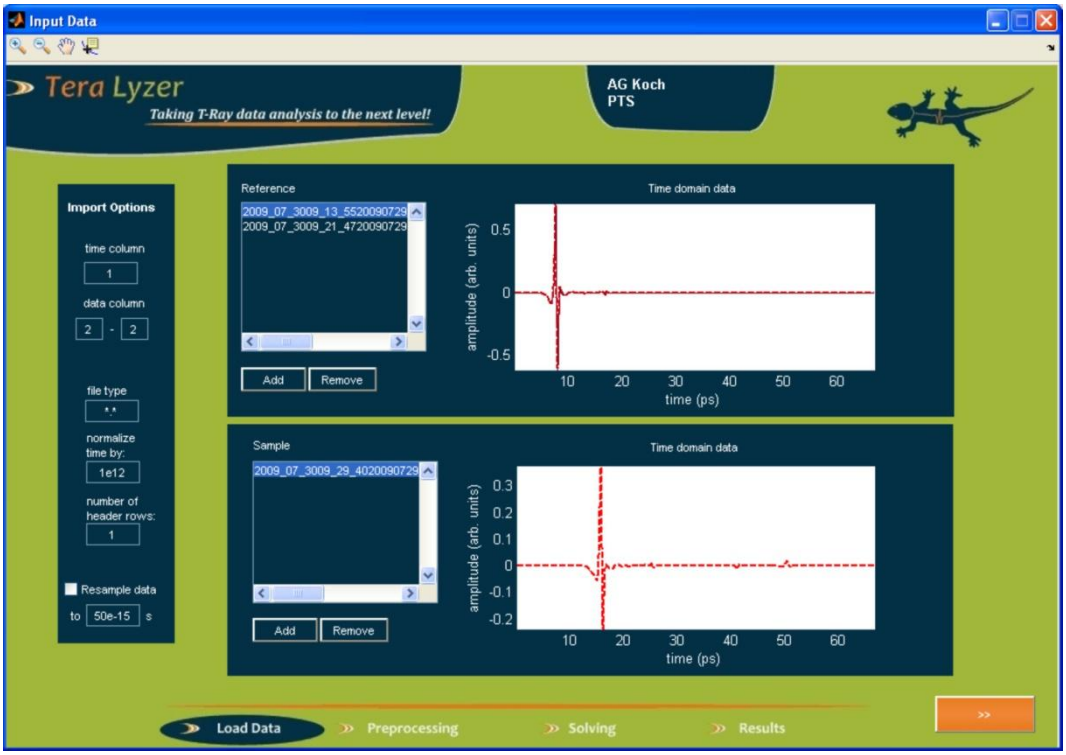
The right hand graph shows the Quasi space peak value for different thickness. At the minimum of the V-like curve lies the correct sample thickness at which the Fabry Perot oszillations on the frequency domain material parameters are minimized. Optionally, error bars can be displayed by marking the corresponding checkbox. As a final step, the complete dataset can be saved as a comma separated ASCII file (.csv) or as Matlab® file (.mat).

4.2 Extracting material parameters of a liquid crystal inside a cuvette

Extracting the THz material parameters of liquids is a challenging task as it usually comprises measurements in a cuvette system. The TeraLyzer is able to determine the material parameters of a single unknown layer in a multilayer system. As an example, in the following we will extract the material parameters of a liquid crystal.

4.2.1 Step I - Loading the time domain data

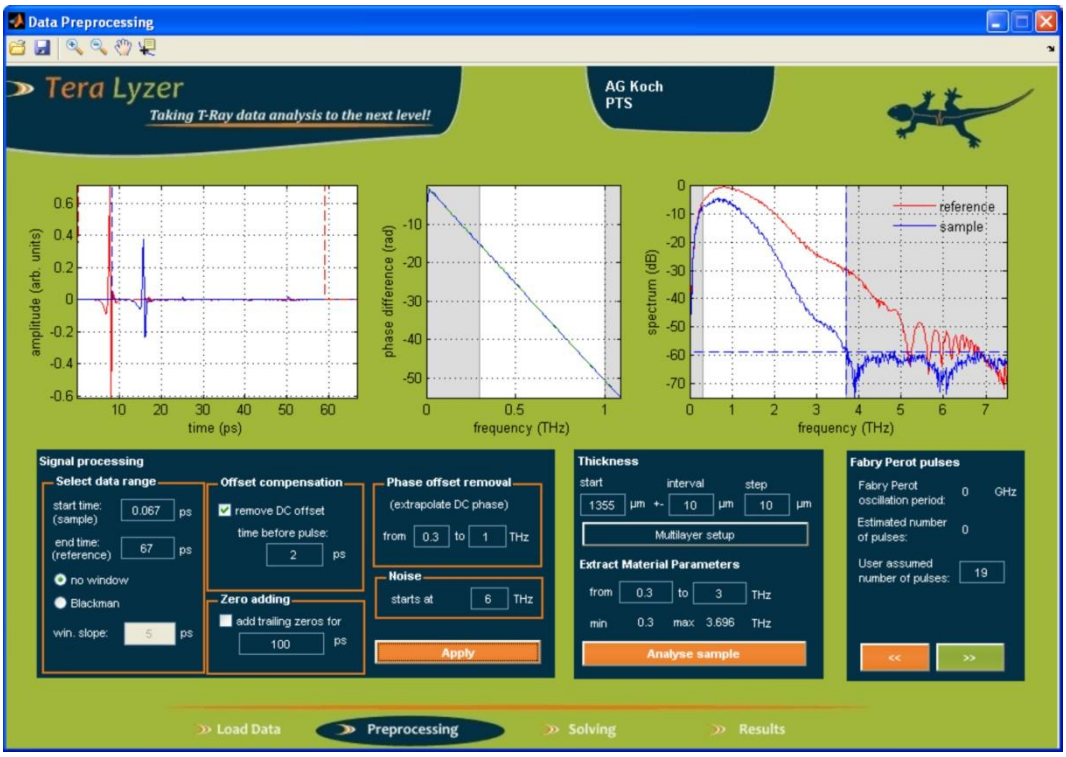
Import your recorded THz time domain pulses from an ASCII file into the TeraLyzer. The import window helps you to find the correct import settings. For advanced statistics, you can load multiple data sets.



In our case, the ASCII file's first column contains the time in ps and the second tab-separated column contains the corresponding THz data. Thus we set the "time column" to 1 and the "data column" to 2. As the time axis has the unit ps we set the "normalize time by" field to 1e12 to obtain the time axis in seconds. Our data has no header rows so that we leave the "header row" option 0.

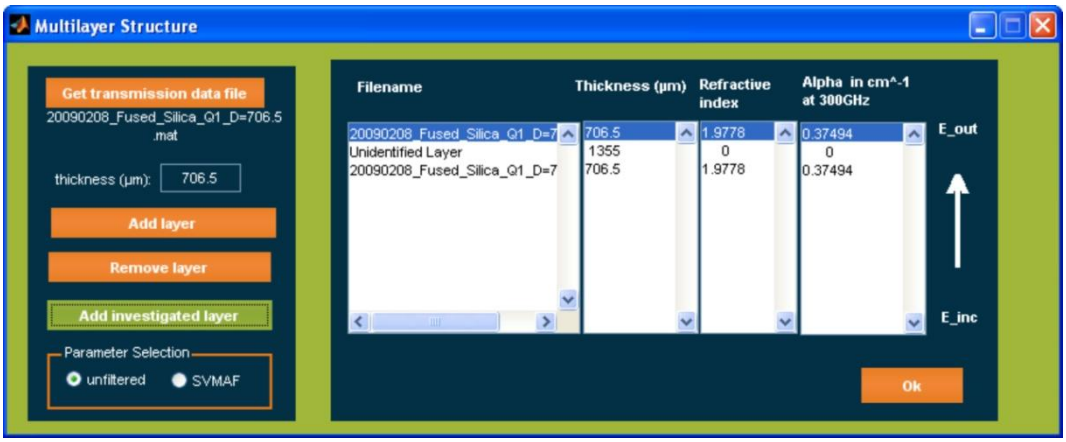
4.2.2 Step II - Preprocessing and Multilayer Setup

The next window offers many preprocessing options like DC offset removal, time-domain window in, phase correction and appending zeros. Furthermore, the frequency range for which parameters should be extracted can be determined here.

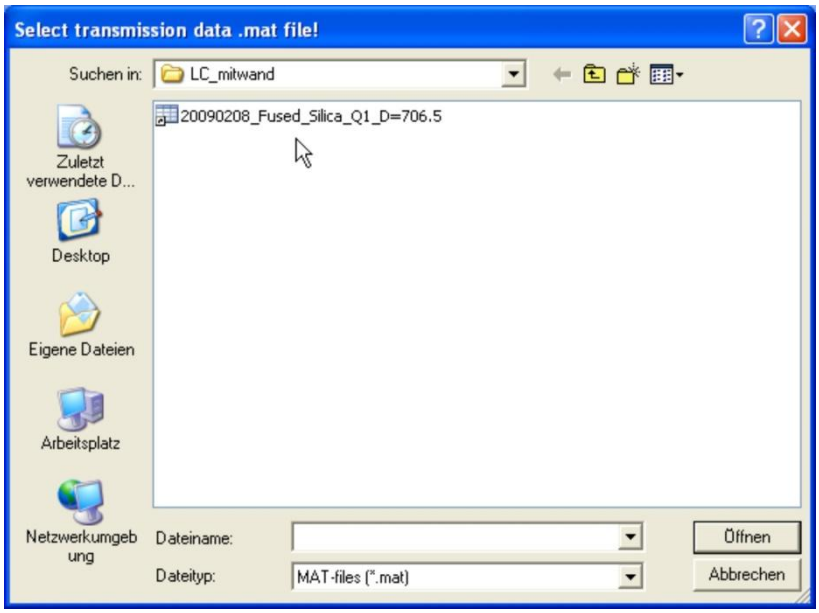


In our case, we select the full available data range from 0 to 66.65 ps. The DC offset is removed by averaging over the first 2 ps of the time domain data and subtracting this average from the data set. We select the zero adding function to artificially increase our frequency resolution, which is legitimate as the time signals at the end of the measured pulse have decreased to the noise level. As phase errors may occur at very low frequencies due to non-planar phase fronts and strong diffraction, we select a trusted phase regime from 0.3 to 0.8 THz. From the spectral data, we see that the noise starts at roughly 4 THz, so we chose this value and enter it into the program. By hitting the "Apply" button, the changes take place.

Now, we enter the multilayer setup by pressing the "Multilayer setup" button in the center panel. The following window opens:



Previous to the cuvette data extraction, we characterized the single side walls of the cuvette with the TeraLyzer. This single layer evaluation is similar to the one performed in our first example, the alpha lactose compound. We saved that data as .mat file. Now, we click the "Get transmission data file" button to select this file.



Please note, that a separate analysis of each layer except the one to be characterized is essential before the multilayer analysis can be perfomed. A scan with an empty cuvette does not suffice!

The thickness of the side wall is stored in the .mat file. However, if a different thickness is required due to a changed setup you can manually enter it into the "thickness" edit box. You can create the multilayer system that you measured by adding layers from transmission files (using the "Add layer" button) and by adding an unidentified layer that you would like to characterize (by pressing the "Add investigated layer" button). In our case we enter the cuvette scenario with the unidentified layer in the middle surrounded by the cuvette walls. When you are finished with the multilayer setup, press "Ok".

Back in the preprovessing widow, we chose the sample thickness to be within 1355 µm +-10 µm. The material parameter extraction frequency range is set to 0.5 THz to 1.6 THz. By hitting the "Analyze Sample" button, an initial evaluation is performed and the position of the Fabry Perot echoes is displayed in the time domain data plot. We see that 3 Fabry Perot echoes lie within our sample range. To procede we hit the ">>" button.

4.2.3 Step III - Numerical Material Parameter Extraction

After the Preprocessing is performed, the calculation of the parameters takes place. Depending on the thickness iteration steps, this process might take a few minutes.

In the following window, the material parameters are displayed.

4.2.4 Step IV - Material Parameters Results

After the calculation is finished, the resulting material parameters are displayed. The dispersion characteristics of the liquid crystal which was analyzed are nicely reproduced with high accuracy.

The right hand graph shows the Quasi space peak value for different thickness. At the minimum of the V-like curve lies the correct sample thickness at which the Fabry Perot oszillations on the frequency domain material parameters are minimized. Optionally, error bars can be displayed by marking the corresponding checkbox. As a final step, the complete dataset can be saved as a comma separated ASCII file (.csv) or as Matlab® file (.mat).